

Cheap Scores First: Cascading Ablated Auxiliary Models for Targeted Data Selection

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Abstract

Pretraining data selection and curation are critical steps for training high-quality LLMs. Model-aware data selection methods such as CoLoR-Filter (Brandfonbrener et al., 2024) use a pair of small auxiliary LLMs to score and select pretraining data. Under compute constraints, we investigate the trade-offs in transformer-block deletion ablations of 150M-class auxiliary models. Ablating the middle two blocks leaves CoLoR scores largely intact, with a Spearman rank correlation of 0.89 against the full-model scores on a sampled C4 pool, whereas top- and bottom-two deletions degrade the correlation to 0.53 and 0.49. However, around selection thresholds where data decisions are actually made, we find that the scored ablations collapse to near chance. To mitigate this, we propose a cascading approach: score samples cheaply with an ablated model, followed by final filtering with the full model. Our proposed cascading approach improves AUC from 0.51 to 0.68 around the hardest boundary threshold. As an end-to-end evaluation, we train 410M-class models on cascade-selected data and find Books validation loss to be competitive with the baseline (5.004 vs. 5.038 cross-entropy loss, averaged over three training seeds) despite selecting substantially different data (0.49 Jaccard overlap). These results suggest auxiliary-model depth is a practical lever for cheaper targeted data selection.

1 Introduction

Data selection for training large language models remains challenging, especially under compute constraints (Longpre et al., 2023; Albalak et al., 2024). Although frontier LLMs are colloquially said to be trained on “the entire internet,” in practice heavy filtering is applied (Raffel et al., 2020; Rae et al., 2021; Penedo et al., 2023; 2024). Besides manual curation of training data (Gao et al., 2020; Soldaini et al., 2024), there is considerable interest in automating data selection. Automated methods can range from model-agnostic techniques, such as deduplication (Lee et al., 2022; Abbas et al., 2023), to model-aware techniques that use auxiliary models to score data with respect to a target distribution (Mindermann et al., 2022; Xie et al., 2023; Fang et al., 2023; Engstrom et al., 2024; Wettig et al., 2024). One such model-aware approach is CoLoR-Filter (Conditional Loss Reduction Filtering) (Brandfonbrener et al., 2024): the difference between the log-likelihoods of two small auxiliary models (a pre-trained “marginal” and a downstream fine-tuned “conditional” model) is used to select sequences for training significantly larger models. The cost of data selection is then dominated by these auxiliary models, so understanding how small they can be made is especially beneficial under computational constraints.

In this paper, we ask how much of the auxiliary models can be removed before selection quality degrades. We use the auxiliary models and filtering scheme introduced by Brandfonbrener et al. (2024). We first ablate the 150M-class auxiliary models used to compute CoLoR scores¹, deleting transformer blocks from the conditional and marginal models,

¹Each full auxiliary scorer has 151M non-embedding parameters, or 254M total parameters including token embeddings and the untied output head. Each trained target model has 393M non-embedding parameters, or 522M total parameters.

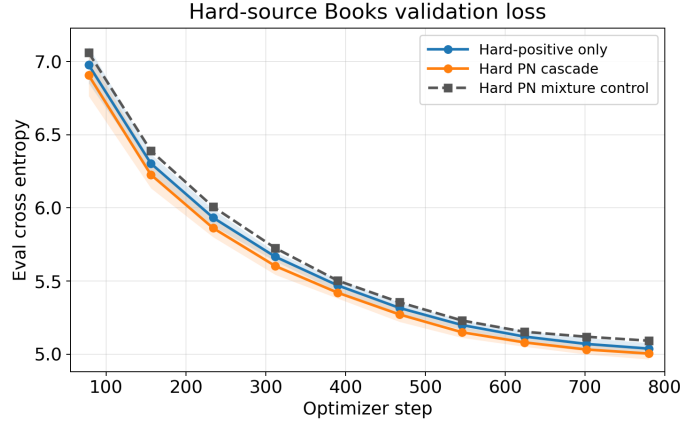


Figure 1: Books validation loss for 410M models trained on 100K sequences selected near the decision boundary, the hardest setting we test. Our two-stage cascade (orange) selects with a cheap ablated scorer plus full-model reranking. Hard-positive only and Hard PN cascade are averaged over three training seeds with shaded ± 1 s.d.; Hard PN mixture control is a single-seed control. Despite choosing substantially different data than full-model selection (0.49 Jaccard), the cascade reaches lower final loss than Hard-positive only (5.004 vs. 5.038) and the mixture control (5.092).

and find that ablated auxiliary models degrade unevenly: they retain much of the full scorer’s ranking signal on well-separated data but collapse to near chance precisely at the selection boundary. Motivated by this, we propose a cascading selection scheme: a cheap ablated auxiliary pair performs a fast first pass that discards clearly irrelevant data, and the full auxiliary pair rescores only the surviving candidates. Finally, we validate the cascade end-to-end by pretraining 410M-class target models on the selected data, showing that cascaded selection recovers most of the downstream performance of full-model CoLoR selection at reduced scoring cost.

Our contributions are:

- **A robustness study of CoLoR scoring under auxiliary-model block ablations.** Deleting middle blocks preserves CoLoR score rankings (0.89 Spearman vs. full-model scores); deleting top-two or bottom-two blocks does not (0.53 and 0.49). But near the selection boundary, the scored ablations collapse to chance (Section 3).
- **A two-stage cascade that uses the ablated pair as a prefilter.** Because reranking a candidate set requires only enough ranking fidelity to preserve candidates, a middle-block-ablated first pass followed by full-model reranking of $1.5\times$ the budget lifts boundary-local AUC from 0.51 to 0.68 and improves every pairwise task (Section 4).
- **An end-to-end validation at 410M scale.** On the hardest boundary-local Books setting, models trained on cascade-selected data improve over full-model selection in seed-averaged validation loss (5.004 vs. 5.038 over three seeds) and in each paired seed, despite selecting substantially different data (0.49 Jaccard overlap; Section 5)

2 Background: CoLoR-Filter algorithm

CoLoR-Filter, introduced by Brandfonbrener et al. (2024), first trains two small auxiliary models. The first auxiliary model is the “marginal” model, a small LLM trained on prior training data. The second auxiliary model is the “conditional” model obtained by fine-tuning the marginal model on the downstream task data. Given these two auxiliary models, we then compute log-likelihood difference scores for each data point $c(x) = -\log p_{\theta_{\text{cond}}}(x) + \log p_{\theta_{\text{marg}}}(x)$. See Appendix Algorithm 1 for further details.

Brandfonbrener et al. (2024) show that 150M-class auxiliaries select data that improves target models up to 1.2B parameters, on both domain adaptation (selecting from C4 for Books) and a suite of multiple-choice QA tasks. Nevertheless, computing $c(x)$ still requires forward passes of *both* full auxiliary models over the *entire* candidate pool. Scoring therefore dominates the cost of selection, which motivates our central question: how much of θ_{marg} and θ_{cond} can be removed while preserving the quality of the selected set?

3 Auxiliary Model Ablations

Motivated by evidence that many transformer blocks can be removed with minimal performance loss (Gromov et al., 2025; Men et al., 2024), we investigate ablating the already trained auxiliary models θ_{marg} and θ_{cond} provided by Brandfonbrener et al. (2024). We consider transformer-block deletion ablations of the CoLoR auxiliary scorer; Appendix Table 1 defines the ablations we considered.

To evaluate the ablations, we test whether these scorers recover examples from the Books (Soldaini et al., 2024) selection task studied by Brandfonbrener et al. (2024). We use the released color-filtered-c4-books subset, which contains C4-derived documents already selected by the original Books CoLoR-Filter pipeline as Books-like training data, and sample 5,000 of these pre-filtered Books-selected sequences. We find middle-block deletion is robust: pair_mid2 (deletion of the middle two blocks) retains 95.8% and pair_mid4 (deletion of the middle four blocks) retains 92.5% of these Books-selected diagnostic sequences at the $\tau = 16$ C4 selection threshold (the original pipeline selects at $\tau = 64$ (Brandfonbrener et al., 2024); we use $\tau = 16$ for our screening diagnostics). See Appendix Table 2 for metrics for other ablations.

To further understand ablations, we also construct a pool from a shuffled stream of C4 (Raffel et al., 2020) English training documents, packed into 100K sequences of length 512. In addition to scoring this C4 pool with each ablated model, we also compute full-model scores as a baseline. As we vary the selection thresholds used in Brandfonbrener et al. (2024), the recall of pair_mid2 remains close to 80% (Appendix Figure 2). The pair_mid2 variant, which deletes the middle two blocks, also retains 0.89 Spearman correlation with the full-model scores (Appendix Table 2). pair_top2 and pair_bot2 are much more destructive, dropping Spearman correlation to 0.53 and 0.49, respectively.

Our finding that middle blocks in LLMs are more robust against perturbations is consistent with earlier findings (Gromov et al., 2025; Lad et al., 2024). However, we suspect the score perturbations induced by ablations are largest around the decision threshold. To test this, we construct data pools around the data-selection threshold (see Appendix Figure 3 for the CDF around the tail of the score distribution).

3.1 Score-pool construction

Uniform C4 samples make the ablated scorers look stronger than they are, because most documents are far from the Books selection boundary. CoLoR-Filter keeps the lowest-scoring $1/\tau$ fraction of a candidate pool; at the operating point of Brandfonbrener et al. (2024), $\tau = 64$, this is only the lowest 1.56% of the score distribution. Appendix Figure 3 zooms into this lower tail and shows the nearby $\tau \in \{16, 32, 64\}$ cutoffs.

We therefore evaluate ablations on boundary-local score pools. Let S_{64} be the full-scorer selection at $\tau = 64$, and let S_{32} be the larger $\tau = 32$ selection. We treat sequences from S_{64} as “positives” and those outside as “negatives”; hard negatives come from the nearby band $S_{32} \setminus S_{64}$; tail negatives come from outside S_{64} . We sample 100K rows per pool and form balanced pairwise tasks from these pools. Appendix Table 3 gives the exact sampling rules.

Macro-averaging over the six pairwise tasks, the full scorer reaches ROC AUC 0.9679, AP 0.9670, and Spearman 0.8969; among cheap variants, pair_mid2 is strongest (0.8079 ROC AUC, 0.8151 AP, 0.5954 Spearman; Appendix Table 4). The per-task breakdown (Appendix Tables 5–10) shows the same pattern: pair_mid2 is the strongest ablated variant on every

pairwise task, with the reported score-pool ablations struggling most on the hard-positive versus hard-negative boundary task and performing best when tail negatives are included.

Hard-positive versus hard-negative comparisons test local fidelity at the decision boundary, which serves as a hard diagnostic for the ranking of ablated models. This comparison is difficult even for the full scorer: because pool labels derive from the officially released scores from Brandfonbrener et al. (2024) while our rescoring is an independent bf16 reproduction, the full baseline reaches only 0.82 AUC rather than the ideal 1.0. Against this ceiling, the reported score-pool ablations — including middle-block deletions — collapse to near-chance discrimination (AUC and AP near 0.5).

4 Cascading

To improve the fidelity of ablated auxiliary models, we propose a two-stage cascading approach. In the first stage we keep a small candidate set using ablated models, then rerank only that set with the full model. Specifically, given a final selection budget of k sequences and a candidate multiplier $m_c > 1$, the cascade scores every row with the cheap scorer $\tilde{c}$, keeps the $\lceil m_c k \rceil$ lowest-scoring rows as candidates, reranks only those candidates with the full scorer c , and returns the best k rows (Appendix Algorithm 2). The multiplier m_c controls the safety margin: a larger candidate set costs more full-model rescoring but tolerates more ranking error in the cheap first pass. Here, we use the ablated pair_mid2 for the cheap scores (middle two blocks deleted from both the marginal and conditional auxiliary models). We use $m_c = 1.5$ throughout.

The cascade improves AUC over the ablated pair_mid2 scorer across all six pairwise score-pool tasks (Appendix Table 11). In the hardest boundary-local comparison (hard positives versus hard negatives), we see an AUC improvement from 0.51 to 0.68.

5 End-to-end training evaluation

Final utility depends on downstream training, so we use the domain-transfer setup from Brandfonbrener et al. (2024): train 410M-class target models on selected C4 data and evaluate Books cross-entropy loss (Appendix F). In the hard-source boundary-local comparison highlighted in Figure 1, the cascade improves Books validation loss over the positive-only baseline (5.004 versus 5.038 averaged across three training seeds), with lower Books loss in each paired seed, and over the single-seed Hard PN mixture control (5.092). On the single-seed random-source mini-universe, the cascade ends close to the positive-only baseline (4.873 versus 4.839) and below the mixture control (4.945; Appendix Figure 4). These 410M-class runs are not a scaling claim, but they show that the cascade can remain competitive after actual target-model training.

6 Conclusion

CoLoR-Filter, introduced by Brandfonbrener et al. (2024), provides an efficient method for model-aware data selection. We show it can be made cheaper still by deleting middle blocks from the auxiliary models. However, near the selection boundary, where selection decisions are actually made, an ablated scorer’s ranking degrades to chance. A two-stage cascade between the cheap ablated scorer and the full model recovers much of this degradation, raising boundary-local ROC AUC from 0.51 to 0.68, matching or slightly improving on full-model selection in seed-averaged validation loss at 410M scale, and reducing scoring cost by an idealized 14.3% (the ablated scoring pass alone measured $0.88 \times$ full-scorer wall-clock; Appendix E.1). A natural next step is multi-stage cascades in which even deeper ablations handle the earliest filtering stages.

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228 A CoLoR-Filter Data Selection

229 CoLoR-Filter uses two auxiliary language models to rank candidate training sequences
 230 for a downstream target distribution. A marginal model is trained on prior data, and a
 231 conditional model is initialized from it and fine-tuned on downstream examples; the CoLoR
 232 score is the conditional loss minus the marginal loss, so lower scores indicate examples that
 233 look more selectively useful for the downstream task. Algorithm 1 reproduces the original
 234 selection procedure from Brandfonbrener et al. (2024); our experiments keep this scoring
 235 convention but ablate or cascade the auxiliary scorer.

Algorithm 1 CoLoR-Filter data selection (from Brandfonbrener et al. (2024)).

Inputs: prior corpus D_{prior} , downstream corpus D_{down} , candidate training corpus D_{train} ,
 selection budget n , and subset multiplier τ .

1. Train a marginal auxiliary model θ_{marg} on D_{prior} .
 2. Initialize a conditional auxiliary model from θ_{marg} and fine-tune it on D_{down} to
 obtain θ_{cond} .
 - 236 3. Draw a candidate pool $D_{\tau} \subset D_{\text{train}}$ with $|D_{\tau}| = \tau n$.
 4. For each $x \in D_{\tau}$, compute

$$c(x) = -\log p_{\theta_{\text{cond}}}(x) + \log p_{\theta_{\text{marg}}}(x).$$
 5. Return the n examples in D_{τ} with the smallest values of $c(x)$ as the selected set
 S for target-model training.
-

237 B Layer-Ablation Screening

238 B.1 Ablation variants

239 Table 1 defines the paired ablation variants referenced in the paper’s layer-screening and
 240 score-pool robustness studies. The full baseline is included as the unablated reference scorer.
 241 Layer indices use bottom-to-top indexing from 0 to 11.

Variant	Description
Full baseline	Unablated conditional/marginal reference scorer
pair_top1	Remove original layer 11, the final block, from both models
pair_top2	Remove original layers 10–11, the top two blocks, from both models
pair_bot2	Remove original layers 0–1, the bottom two blocks, from both models
pair_mid2	Remove original layers 5–6, the middle two blocks, from both models
pair_mid4	Remove original layers 4–7, the middle four blocks, from both models

Table 1: Ablation variant definitions.

Variant	Blocks removed	Drop (%)	Recall (1/16)	Spearman ρ	Enriched (1/16)
pair_top1	1	8.333	0.541	0.737	0.853
pair_top2	2	16.667	0.385	0.534	0.721
pair_bot2	2	16.667	0.317	0.485	0.529
pair_mid2	2	16.667	0.755	0.890	0.958
pair_mid4	4	33.333	0.582	0.715	0.925

Table 2: Primary block-ablation results at the 1/16 pure-C4 selection threshold. Recall is measured against the full scorer’s bottom-6,250 pure-C4 selected set. Enriched (1/16) is the fraction of the 5,000 enriched Books-selected diagnostic rows retained when each scorer’s 1/16 pure-C4 threshold is applied to those rows. The table is restricted to paired ablation variants.

242 B.2 Ablation evaluations

243 We first create a frozen evaluation pool designed to test selected-set stability before con-
 244 structing the boundary-local score pools. We stream allenai/c4 English training docu-
 245 ments, shuffle the stream with seed 17 and a 10K-document buffer, tokenize with the
 246 allenai/eleuther-ai-gpt-neox-20b-pii-special tokenizer, append EOS between docu-
 247 ments, and pack the stream into 100,000 contiguous sequences of length 512. The same pool
 248 also includes 5,000 enriched Books-selected sequences, but those enriched rows are excluded
 249 from the overlap and Jaccard denominators; they are used only for a separate tail-retention
 250 diagnostic. Specifically, after the pure-C4 selection threshold is set, we apply that threshold
 251 to the enriched Books rows and report enriched-row recall at the 1/16 operating point: the
 252 fraction of these 5,000 known Books-selected rows that the scorer would keep at the 1/16
 253 pure-C4 threshold.

254 For Figure 2, the reference “ground truth” is not a human label but the selected set pro-
 255 duced by the unablated full conditional/marginal CoLoR scorer, scored in bf16 on an A100.
 256 Since lower CoLoR scores are preferred, at selection rate r we set $k = \lfloor 100,000r \rfloor$ and
 257 compare the full scorer’s bottom- k pure-C4 set with the ablated scorer’s bottom- k pure-C4
 258 set. The plotted value is recall of the full selected set, $|S_{\text{full},k} \cap S_{\text{ablated},k}|/k$, evaluated at
 259 $r \in \{1/64, 1/32, 1/16, 1/8, 1/4\}$. A repeat full-model run (full_rescore) serves as a noise-
 260 floor check. Table 2 summarizes the primary 1/16 operating point, where $k = 6,250$ pure-C4
 261 sequences.

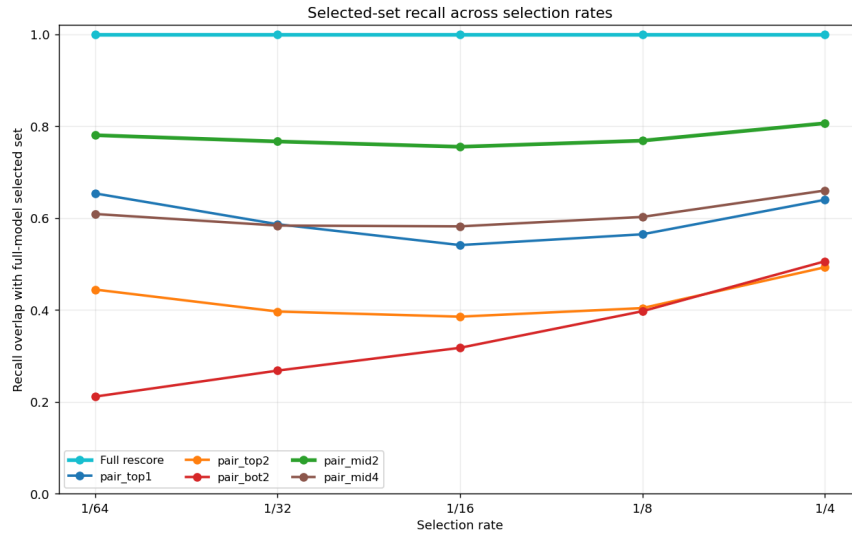


Figure 2: Selected-set recall of block-ablated CoLoR scorers relative to the full scorer across selection rates on the fixed C4 evaluation pool. Each point reports the fraction of the full scorer’s selected set recovered by the ablated variant at the same selection rate. The pair_mid2 ablation is the most stable cheap scorer among the displayed variants; the repeat full-model rescore is included as the noise-floor reference.

Pool	Source band	Cutoff	Sampling rule
Random positives	S_{64}	$\tau = 64$ boundary	Random 100K rows from S_{64} .
Hard positives	S_{64}	$\tau = 64$ boundary	The 100K rows in S_{64} with the largest CoLoR scores, i.e., positives closest to the $\tau = 64$ boundary from below.
Random negatives	$S_{32} \setminus S_{64}$	$\tau = 32$ boundary	Random 100K rows from the band selected at $\tau = 32$ but not at $\tau = 64$.
Hard negatives	$S_{32} \setminus S_{64}$	$\tau = 32$ boundary	The 100K rows in $S_{32} \setminus S_{64}$ with the smallest CoLoR scores, i.e., negatives closest to the $\tau = 64$ boundary from above.
Tail negatives	All scored rows outside S_{64}	outside S_{64}	Random 100K rows from all scored rows outside S_{64} .

Table 3: Definition of the five score pools used for robustness and cascade analysis. Hard positives and hard negatives form a boundary-local stress test, whereas random positives and tail negatives give easier, more separated comparisons.

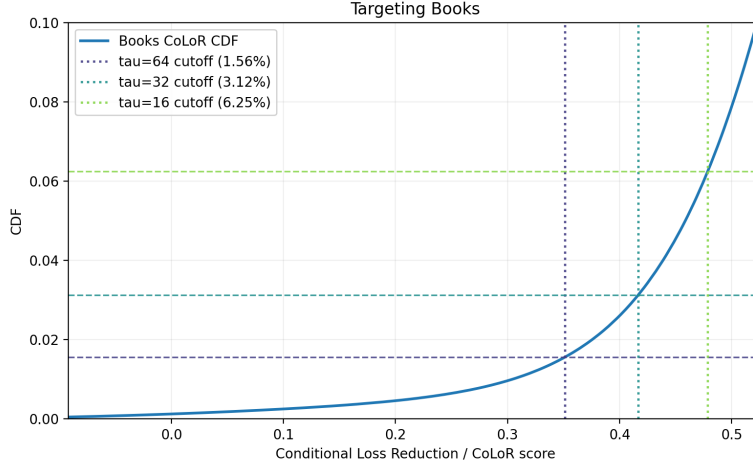


Figure 3: Books CoLoR score CDF defining the score pools. S_{64} is the lowest-scoring 1/64 of C4 rows (positives); $S_{32} \setminus S_{64}$ is the adjacent band used for hard negatives; $\tau = 16$ gives broader-tail context.

262 C Score-Pool Construction

263 The score pools turn the continuous full-model Books CoLoR ranking into controlled binary
 264 tasks for measuring whether an ablated scorer preserves the same ordering. They are built
 265 from the released CoLoR selected-index artifacts, not from manual labels: for a subset
 266 multiplier τ , the full scorer keeps the lowest-scoring $1/\tau$ fraction of the candidate pool.
 267 Thus S_{64} is the selected set at the paper operating point, and S_{32} is a larger selected set that
 268 contains S_{64} plus nearby examples just above the $\tau = 64$ boundary.

269 We use these nested sets to separate easy ranking preservation from the boundary-local case
 270 that matters for selection. Random positives and random negatives sample broadly within
 271 their source bands, while hard positives and hard negatives are chosen closest to the $\tau = 64$
 272 boundary, where small score perturbations can change membership in the selected set. Tail
 273 negatives are sampled from outside S_{64} and give the most separated negative comparison.
 274 Each pool contains 100K rows, giving 200K scored examples in each balanced pairwise
 275 robustness task. Table 3 gives the exact sampling rules, and Figure 3 visualizes how the
 276 $\tau \in \{64, 32, 16\}$ cutoffs sit in the compressed lower tail of the Books score distribution.

Variant	Ablation	ROC AUC	AP	Spearman
Full baseline	none	0.9679	0.9670	0.8969
pair_mid2	remove middle two blocks from both scorers	0.8079	0.8151	0.5954
pair_top1	remove final block from both scorers	0.7751	0.7799	0.5319
pair_mid4	remove middle four blocks from both scorers	0.7675	0.7720	0.5170
pair_top2	remove top two blocks from both scorers	0.7412	0.7406	0.4677

Table 4: Primary score-pool robustness results, macro-averaged over the six pairwise pool comparisons. The ablation column summarizes which layer blocks are removed from the conditional and marginal scorers. Higher ROC AUC, AP, and Spearman correlation indicate better preservation of the full-model ranking signal.

D Score-Pool Robustness Results

D.1 Aggregate score-pool robustness

Table 4 macro-averages over the six balanced pairwise score-pool tasks. Higher ROC AUC, average precision, and Spearman correlation indicate better preservation of the full-model ranking signal. The official score-pool scoring run used for Tables 4–10 did not include pair_bot2, so the bottom-two result is reported only for the layer-screening sweep in Table 2 and Figure 2.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	0.8159	0.8113	0.6249
pair_mid2	paired	0.5098	0.5073	0.0215
pair_top1	paired	0.5066	0.5041	0.0134
pair_mid4	paired	0.5063	0.5039	0.0131
pair_top2	paired	0.5047	0.5027	0.0100

Table 5: Per-task score-pool metrics for hp_vs_hn: hard positive versus hard negative.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	0.9956	0.9939	0.9181
pair_mid2	paired	0.6792	0.6618	0.3449
pair_top1	paired	0.6189	0.5975	0.2270
pair_mid4	paired	0.6116	0.5929	0.2174
pair_top2	paired	0.5896	0.5691	0.1715

Table 6: Per-task score-pool metrics for hp_vs_rn: hard positive versus random negative.

284 D.2 Per-task score-pool robustness

285 Tables 5–10 report the AUC, average precision, and Spearman correlation separately for each
 286 positive/negative pool comparison. Each task has 100,000 positives and 100,000 negatives.
 287 Bold entries mark the best ablated variant in each metric; the full baseline is shown for
 288 reference but excluded from winner marking. All reported ablations in these tables use
 289 paired conditional/marginal deletion. They are restricted to the variants scored in the
 290 official score-pool run, which did not include pair_bot2.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	0.9958	0.9969	0.9176
pair_mid2	paired	0.8013	0.8430	0.6277
pair_top1	paired	0.7605	0.7984	0.5551
pair_mid4	paired	0.7426	0.7735	0.5222
pair_top2	paired	0.7260	0.7534	0.4912

Table 7: Per-task score-pool metrics for rp_vs_hn: random positive versus hard negative.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	0.9999	0.9999	0.9996
pair_mid2	paired	0.8780	0.9021	0.7669
pair_top1	paired	0.8254	0.8493	0.6708
pair_mid4	paired	0.8070	0.8293	0.6407
pair_top2	paired	0.7825	0.8003	0.5925

Table 8: Per-task score-pool metrics for rp_vs_rn: random positive versus random negative.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	0.9999	0.9999	0.9210
pair_mid2	paired	0.9852	0.9826	0.8506
pair_top1	paired	0.9571	0.9475	0.8012
pair_mid4	paired	0.9588	0.9520	0.7983
pair_top2	paired	0.8942	0.8693	0.6879

Table 9: Per-task score-pool metrics for hp_vs_tn: hard positive versus tail negative.

Variant	Family	AUC	AP	Correlation
Full baseline	baseline	1.0000	1.0000	1.0000
pair_mid2	paired	0.9936	0.9941	0.9607
pair_top1	paired	0.9821	0.9824	0.9238
pair_mid4	paired	0.9787	0.9803	0.9103
pair_top2	paired	0.9502	0.9488	0.8534

Table 10: Per-task score-pool metrics for rp_vs_tn: random positive versus tail negative.

Pairwise task	pair_mid2 AUC	Cascade AUC	Full baseline AUC
hp_vs_hn	0.5098	0.6831	0.8159
hp_vs_rn	0.6792	0.8728	0.9956
hp_vs_tn	0.9852	0.9986	0.9999
rp_vs_hn	0.8013	0.9018	0.9958
rp_vs_rn	0.8780	0.9478	0.9999
rp_vs_tn	0.9936	0.9991	1.0000

Table 11: AUC improvement from the two-stage cascade on balanced 200K-row pairwise score-pool tasks. The cascade uses pair_mid2 as the prefilter with $m_c = 1.5$, then reranks retained candidates with the full CoLoR scorer.

E Cascade Algorithm

The cascade uses the ablated scorer as a high-recall prefilter rather than as the final selector. It first ranks the full pool with the cheap ablated score, keeps a multiple of the final budget, and then applies the unablated full CoLoR scorer only to that smaller candidate set. This makes calibration shifts in the ablated scorer less important than candidate recall: false positives can be removed by reranking, while false negatives are unrecoverable.

Algorithm 2 Two-stage cascade with full-model reranking.

Inputs: candidate pool D , final selection budget k , candidate multiplier m_c , cheap ablated scorer $\tilde{c}$, and full scorer c . Lower scores are preferred.

1. Score every row $x \in D$ with the cheap scorer $\tilde{c}(x)$.
 2. Keep the $\lceil m_c k \rceil$ lowest-scoring rows as candidates C .
 3. Rescore only $x \in C$ with the full scorer $c(x)$.
 4. Return the k lowest-scoring candidates under $c(x)$.
-

Table 11 shows the same idea on the balanced pairwise score-pool tasks at $m_c = 1.5$. The cascade AUC ranks retained candidates by the full CoLoR score and places filtered-out rows below the candidate set, so it captures both the benefit of full-model reranking and the penalty for positives missed by the prefilter. The largest gains occur on the hard boundary tasks, where pair_mid2 alone is weakest.

E.1 Computational cost estimates

Following the cost accounting of Brandfonbrener et al. (2024), CoLoR-Filter has cost $3M_{\text{prior}} + 2\tau n + 3nL$ in auxiliary-model forward-equivalent token units. Here $M_{\text{prior}} = |D_{\text{prior}}|$ is the number of tokens used to train the prior/marginal auxiliary model, n is the selected target-training token budget, and L is the relative forward cost of the target model compared with the auxiliary scorer. The selection-specific term is the two-model scoring pass, $2\tau n$: at $\tau = 64$ this is $128n$, compared with $3nL = 16.5n$ for target-model training when $L = 5.5$.

Block deletion only changes the scoring term. Under an idealized block-proportional model, a paired ablation retaining a fraction ρ of blocks costs $2\rho\tau n$ to score the full pool; for pair_mid2, $\rho = 10/12$, so this is $106.7n$ at $\tau = 64$. The cascade then adds full-model reranking for only $m_c n$ candidates, giving scoring cost

$$2n(\rho\tau + m_c).$$

With pair_mid2 and $m_c = 1.5$, this is $109.7n$, a 14.3% idealized reduction relative to full scoring. The actual wall-clock gain can be smaller because non-block overheads do not scale with ρ ; in our 500K-row simulation, pair_mid2 measured $0.8786 \times$ the full scoring time.

318 Thus the cascade is useful only when the candidate set remains small and the ablated scorer
319 has high candidate recall near the $\tau = 64$ boundary.

F 410M Training Diagnostics

Figures 4–6 provide the supporting diagnostics for the 410M-class training runs in Section 5. We use the domain-transfer setup from Brandfonbrener et al. (2024): selected rows are C4 sequences for target-model training, and the downstream metric is Books validation cross entropy. The base diagnostics compare six OLMo-style target-model runs on matched 100K-row mini-universes: positive-only baselines train on all random-positive or hard-positive rows, PN cascades train on rows selected from the corresponding positive/negative union by the pair_mid2 cascade with multiplier $m_c = 1.5$, and PN mixture controls train on random 100K draws from the corresponding positive/negative unions. The hard-source Books validation plot in Figure 1 additionally repeats Hard-positive only and Hard PN cascade over three training seeds. Base runs use training seed 17; the hard-source repeat runs use seeds 18 and 19 while reusing the matching seed-17 selected training rows. All runs use the same 522M-parameter architecture (393M non-embedding parameters), optimizer and scheduler, tokenizer, sequence length 512, global batch size 256, and evaluation schedule; each run trains for two passes over its selected 100K-row memmap, or 780 optimizer steps.

The cascade-selected data has high overlap with the random positive-only baseline (0.875 Jaccard; 93.3% positive recall) but much lower overlap in the hard boundary-local setting (0.495 Jaccard; 66.2% positive recall), making the hard-source curves and overlap matrix the stricter diagnostic.

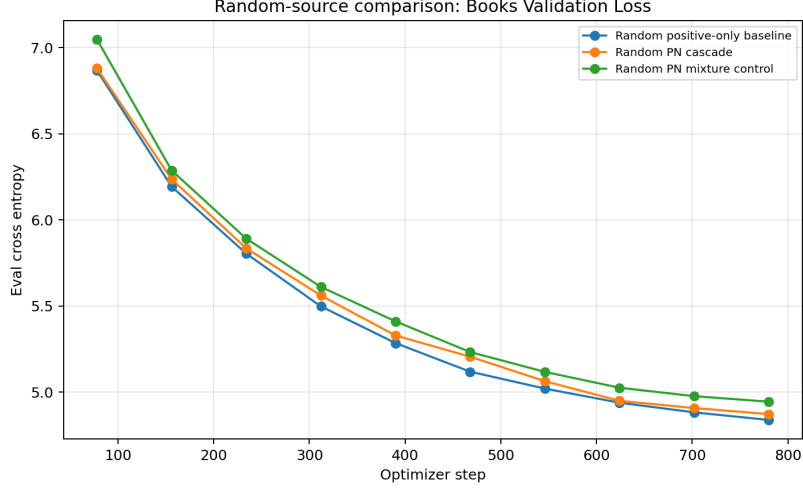


Figure 4: Books validation cross entropy for the random-source mini-universe. Lower is better. The random positive-only baseline is the strongest run, but the random positive/negative cascade remains close (final Books loss 4.873 versus 4.839) and improves over the random positive/negative mixture control (4.945).

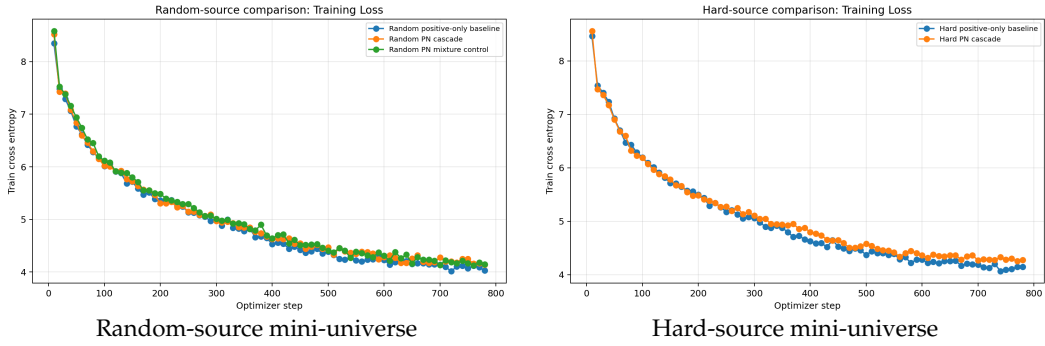


Figure 5: Training cross entropy for the random-source and hard-source mini-universes. These curves verify that the compared target models train under the same schedule and remain close throughout the two-pass run; downstream Books validation loss is reported in Figure 1 and Figure 4. The hard-source panel shows only the seed-17 Hard-positive and Hard PN cascade runs; the hard-source mixture control and seed 18/19 repeats are omitted for clarity.

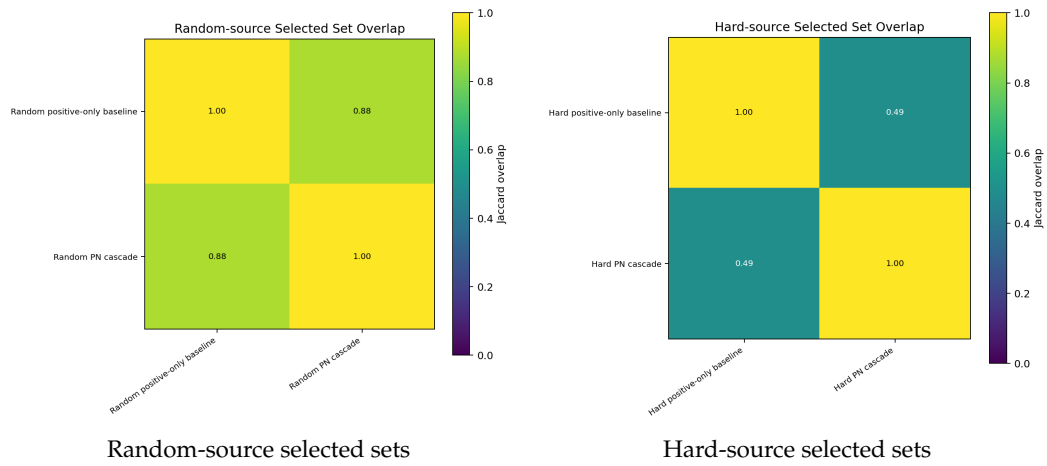


Figure 6: Jaccard overlap between selected training-row sets. The random-source cascade overlaps strongly with the random positive-only baseline (0.88), while the hard-source cascade overlaps much less with the hard positive-only baseline (0.49), reflecting the stricter boundary-local stress test.